

CP Recall

Adaptive DSA Pattern Retention System

We solve many problems.
But do we **retain the patterns**?

**JUST SOLVING
A PROBLEM**

Solve Problem → Remember Solution

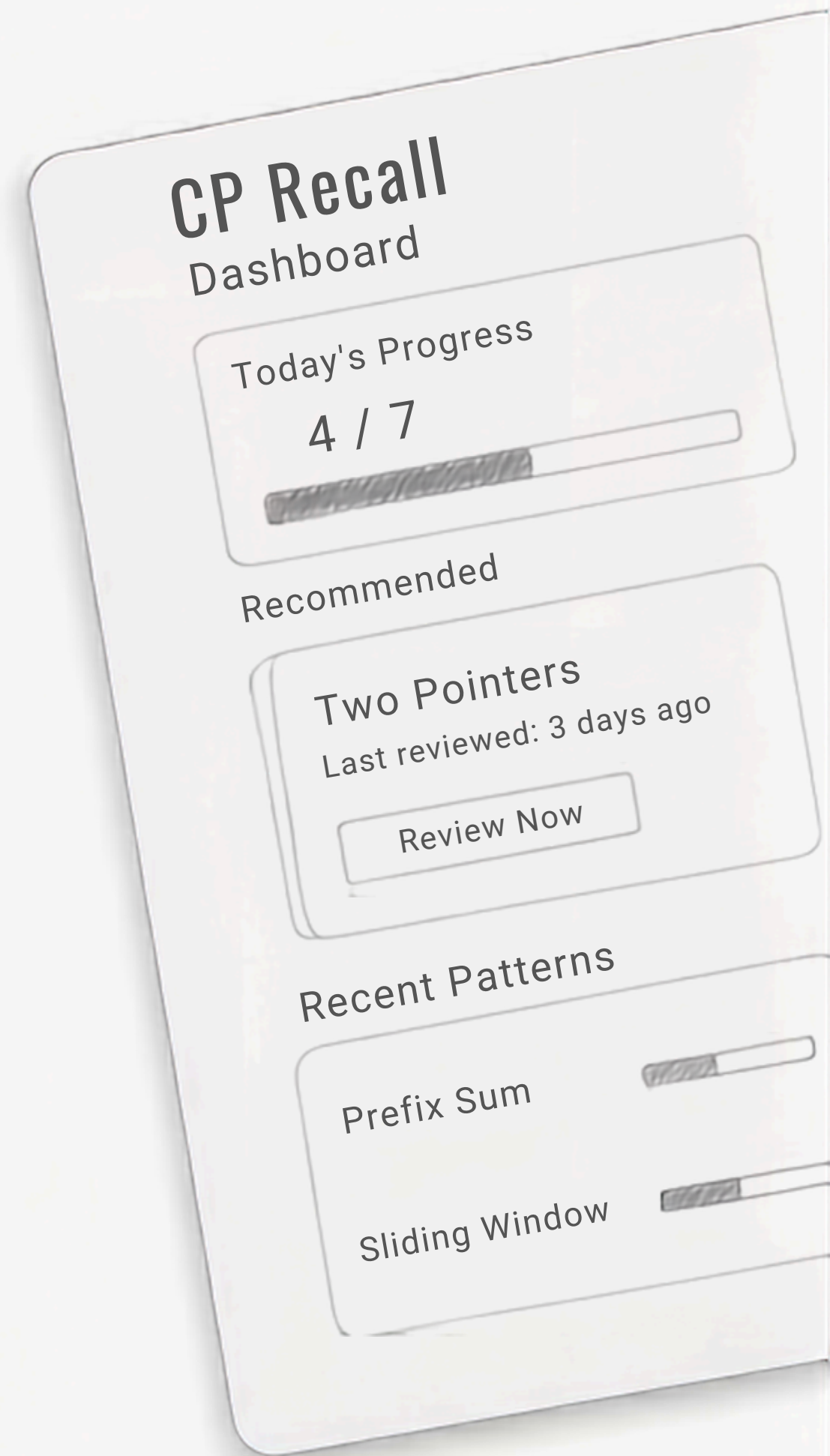


**RETAINING
PATTERNS**

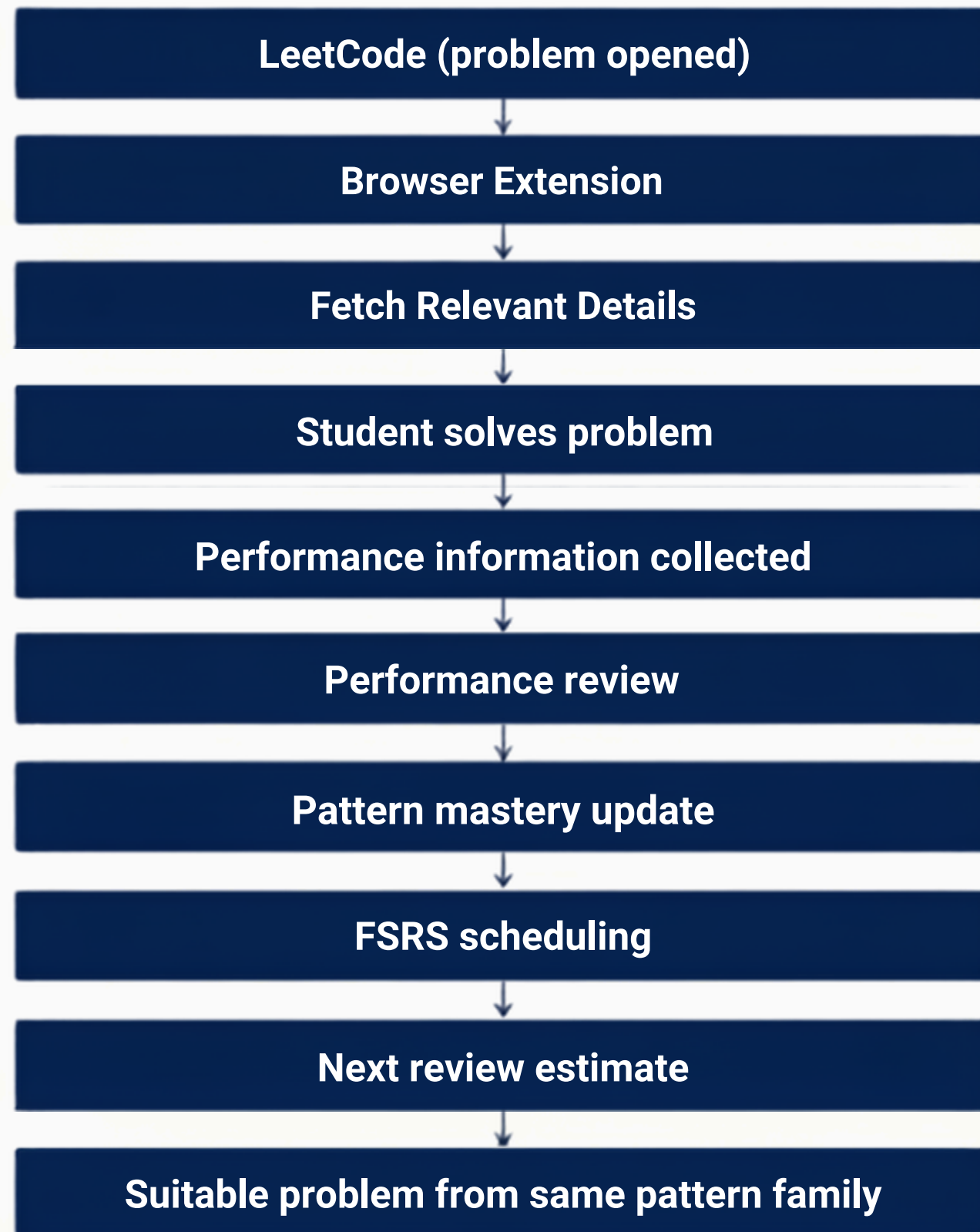
Solve Problem → Get Scored → Solve Different Problem → Learn Pattern



CP Recall focuses on **whether a student can apply a pattern again**,
not just whether they solved a question once.



How CP Recall Works



The extension acts as an interaction layer around the LeetCode workflow; it is not a replacement for LeetCode.

FSRS $\Rightarrow$ **WHEN**

to review

CP Recall $\Rightarrow$ **WHAT**

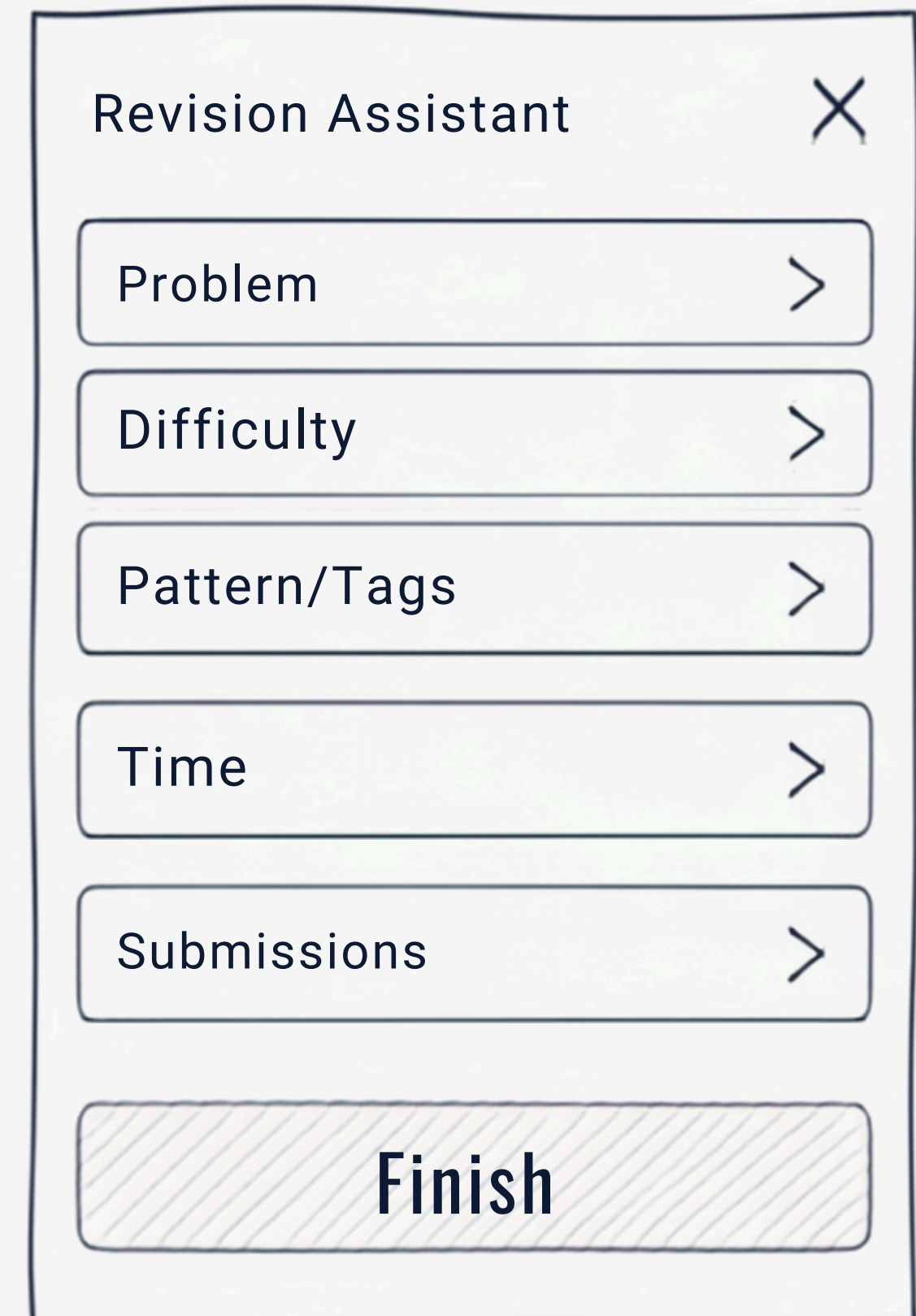
to review / which problem

LeetCode Interaction & Revision Assistant

Student opens a LeetCode problem
Extension detects the current problem
URL
Extension surfaces relevant details
Student solves the problem on LeetCode
Extension fetches the solution data
for further evaluations

LeetCode = **solving environment**

CP Recall extension = **learning/retention assistant**



From Problem Solving to Review Scheduling



What happened on this attempt? Time taken, submissions, result, solution efficiency, time complexity, hints used.

Identify the pattern and track mastery over time.

FSRS uses your past performance to decide when to review again.

Get a clear date for your next review.

After finish

Performance Review	
Time taken	19.35
Submissions	2
Result	✓ Accepted
Solution efficiency	:
Time complexity	:
Hints used	:
[Submit]	

User feedback

STEP 1 – SOLVE

- Student solves the problem on LeetCode.

After submit

Performance Review	
Time Taken	19.35
Submissions	2
Result	76%
Solution efficiency	:
Time complexity	:
Hints used	:
[Submit]	

From Extension + LeetCode

User Input

STEP 2 – PERFORMANCE REVIEW

- Time Taken • Submissions • Result
- Solution Efficiency • Time Complexity
- Hints Used

This information is used to evaluate the student's performance on this attempt.

After submit

Problem	
3 Sum	
Mastery score	76%
FSRS rating	: Good
Next Review	Aug 20, 2026
[Done]	

From Performance Review

Used by + + Used FB.

STEP 3 – PATTERN + FSRS

Problem: 3Sum Pattern: Two Pointers
Mastery: 76% FSRS Rating: Good
Next Review: Aug 20, 2026

The pattern mastery is updated and FSRS determines the next review date.

FSRS RATING SCALE

0 - 39	:	Again
40 - 59	:	Hard
60 - 79	:	Good
80 - 100	:	Easy

"FSRS helps estimate when the pattern should be reviewed again, based on the student's previous performance."

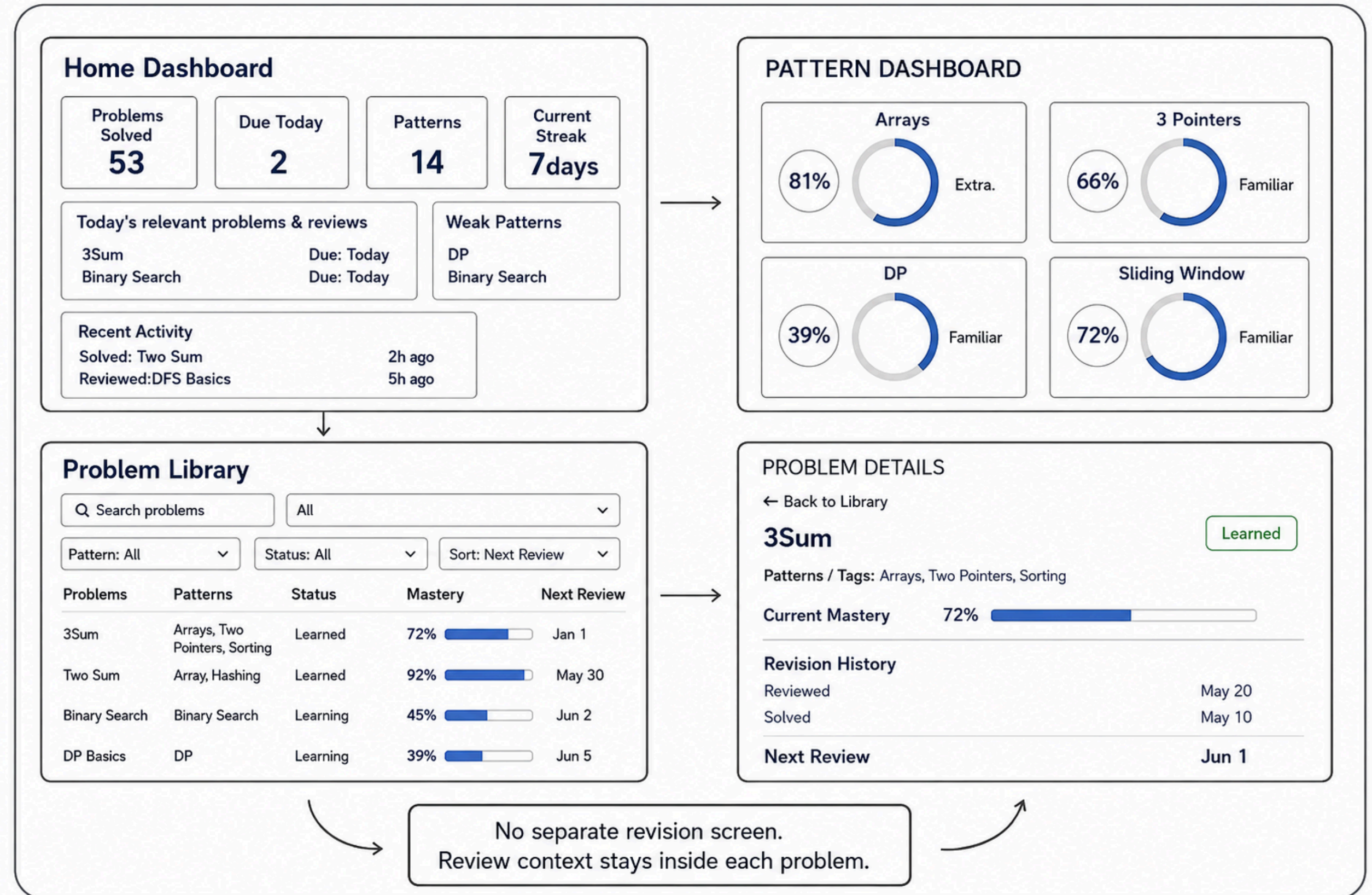
CP Recall Dashboard & Problem Analytics

Every key view **at a glance**:
progress, weak areas, and
what to work on next

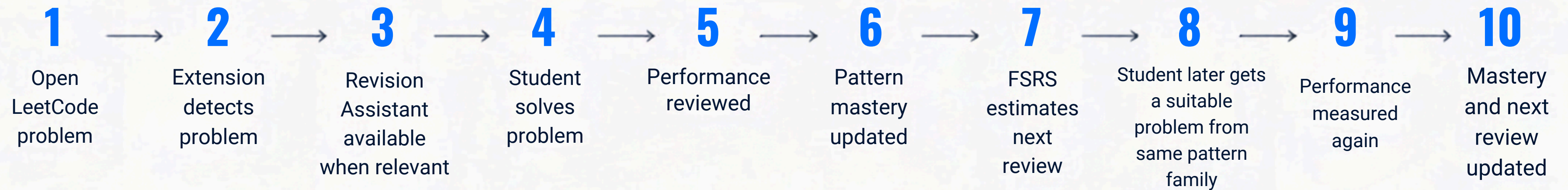
Pattern mastery
highlights **strengths**
and **gaps**

Browse, filter, and sort
problems with **mastery**
and **next review**

Problem details keep
context and history **no**
separate revision page



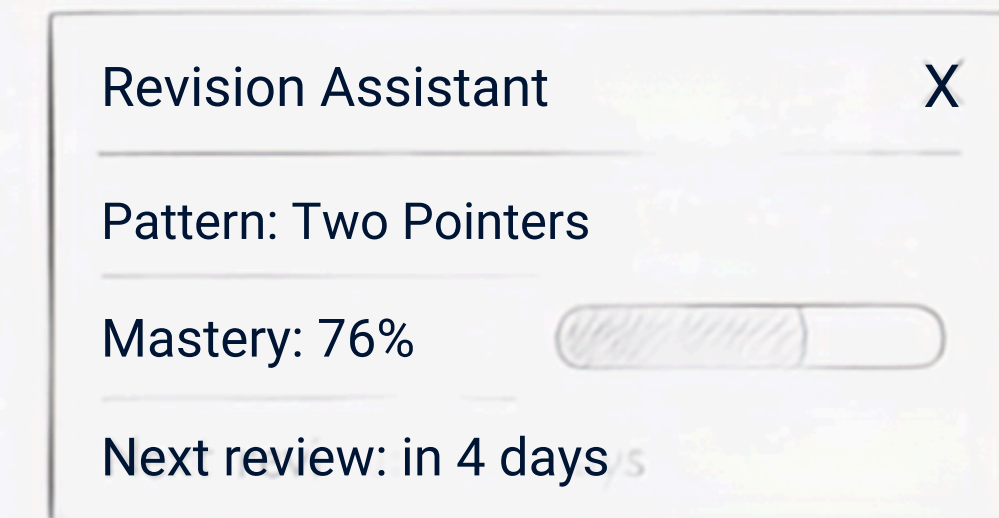
Complete User Journey



Example Flow



Extension Interface (when relevant)



The goal is to test whether the student **retained the problem-solving pattern**, not whether they remember the original question.

Submitted By

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